Siva Satya Vara Prasad Varupala

PhD Student, Industrial and Systems Engineering, Wayne State University, Detroit, MI

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Research Interests

Robotic disassembly and remanufacturing; CAD-based geometric reasoning; learning on 3D geometry (graph neural networks, 3D CNNs on signed distance fields); assembly and disassembly sequence planning; simulation-to-real robot skills.

Education

Ph.D., Industrial and Systems Engineering Aug 2024 – present
Wayne State University. Advisor: Dr. Jeremy L. Rickli. GPA 4.0/4.0 Detroit, MI
Preliminary examinations passed June 2025. Dissertation area: automated (active) disassembly planning.

M.Tech., Mechanical Engineering (Manufacturing Engineering) 2020 – 2022
National Institute of Technology Andhra Pradesh. CGPA 8.98/10, ranked 3rd in class India
Thesis: An automated assembly sequence planner to generate optimal assembly sequence plans through reinforcement learning. Advisor: Dr. M. Hymavathi.

B.Tech., Mechanical Engineering 2014 – 2018
GMR Institute of Technology. CGPA 8.39/10 India
Thesis: Automated assembly sequence generation with subassembly detection. Advisor: Dr. M. V. A. Raju Bahubalendruni.

Honors and Awards

First Place, Intrinsic AI for Industry Challenge (inaugural); Team FLEX, US$100,000 team prize 2026
Finalist, ISIE Hybrid Vehicle Challenge, Imperial Society of Innovative Engineers 2018
Letters of Appreciation (website design; RC car championship coordinator), national student technical contest 2017

Research Experience

Graduate Researcher, Manufacturing and Remanufacturing Systems Laboratory Aug 2024 – present
Wayne State University, with Dr. Jeremy L. Rickli Detroit, MI

- Developed a direction-conditioned 3D CNN that predicts pairwise disassembly interference from two-channel signed distance fields; trained on 3.55 million directional samples from 2,853 assemblies with assembly-disjoint splits. F1 91.04%, and 108.7× faster than CPU gap-stepping analysis (IWAR 2026).
- Co-developed graph neural networks (GAT and GCN) that predict directional interference matrices from CAD assembly graphs for generative design for disassembly (ASME IDETC/CIE 2025; joint first author).
- Built a ROS 2 and Gazebo disassembly prototype that loads STEP assemblies, builds a disassembly network from interference matrices, and recomputes interference after parts settle.
- Built multiprocess CAD data pipelines (STEP cleaning, normalization, signed distance field generation in C++ via pybind11) for datasets of thousands of assemblies.

Intrinsic AI for Industry Challenge, Team FLEX (2 members) 2026
First place. Final decided on a physical UR5e workcell; team ranked first in Phase 1

- Built the team's vision system and both connector-insertion skills for the qualification round.
- Developed the SFP insertion, SC insertion and cable-routing skills for Phase 1.
- For the final, designed a consensus-based task-board detection algorithm and developed the SC and SFP insertion skills that transferred from simulation to the physical workcell, and cable routing with two fallback strategies. The SC skill used a lateral swinging insertion strategy that self-aligns the connector with its port.
- Trained a DAgger-style diffusion policy in NVIDIA Isaac Sim with Newton physics and transferred it to Gazebo.
- Tools: NVIDIA Isaac Sim, Newton, ROS 2, Gazebo, Intrinsic Flowstate, UR5e force control.

Junior Research Fellow May 2019 – Jul 2020
National Institute of Technology Puducherry, with Dr. M. V. A. Raju Bahubalendruni India

- Automated extraction of assembly attributes from CAD models; methods for stable subassembly identification and optimal

linear and parallel assembly sequence planning.

Master's Thesis Research 2020 – 2022
National Institute of Technology Andhra Pradesh India

- Proposed a projection-based method (CSPPA) to build oblique directional interference matrices; prototyped with the FreeCAD Python API and implemented in Open CASCADE, using bounding-box data to limit interference tests on large products. Published in *Computers in Industry*; patent application filed.
- Modified epsilon-greedy Q-learning to generate optimal linear assembly sequences with less computation time than earlier methods.

Publications

Google Scholar (April 2026): 335 citations, h-index 8.

Conference papers

- **S. S. V. P. Varupala**, S. Tinoco-Leon, J. L. Rickli. Predicting disassembly interference matrices from CAD geometry using SDF volumes and CNNs. *IWAR 2026*, Springer LNCS. Accepted.
- **S. S. V. P. Varupala***, J. P. J. Prioli*, J. L. Rickli. Graph neural networks for interference matrix prediction in generative design for disassembly. *ASME IDETC/CIE 2025*, DETC2025-168593. *Equal contribution.
- **S. S. V. P. Varupala** et al. A design for assembly framework based on subassembly detection method. *Recent Advances in Manufacturing Modelling and Optimization*, Springer, 2022, 511–519.
- **S. S. V. P. Varupala**, A. K. Gulivindala, M. V. A. R. Bahubalendruni. An efficient geometrical feasibility testing method for exploded view generation to perform assembly sequence planning. *Current Advances in Mechanical Engineering*, Springer, 2021, 963–973.

Journal articles

- **S. S. V. P. Varupala**, M. Hymavathi, C. S. P. Rao, M. V. A. R. Bahubalendruni. A novel computative strategic planning projections algorithm (CSPPA) to generate oblique directional interference matrix for different applications in computer-aided design. *Computers in Industry*, 141, 103703, 2022.
- G. A. Kumar, M. V. A. R. Bahubalendruni, **S. S. V. P. Varupala**, D. Ashok, K. Sankaranarayanasamy. A novel geometric feasibility method to perform assembly sequence planning through oblique orientations. *Engineering Science and Technology, an International Journal*, 26, 100994, 2022.
- M. V. A. R. Bahubalendruni, **S. S. V. P. Varupala**. Disassembly sequence planning for safe disposal of end-of-life waste electric and electronic equipment. *National Academy Science Letters*, 44(3), 243–247, 2021.
- A. K. Gulivindala, M. V. A. R. Bahubalendruni, A. K. Inkulu, **S. S. V. P. Varupala**, K. Sankaranarayanasamy. A modified cut-set method for mechanical subassembly identification. *Assembly Automation*, 2021.
- G. A. Kumar, M. V. A. R. Bahubalendruni, **S. S. V. P. Varupala**, K. Sankaranarayanasamy. A multi-layered disassembly sequence planning method to support decision making in de-manufacturing. *Sādhanā*, 46(2), 2021.
- A. K. Gulivindala, M. V. A. R. Bahubalendruni, **S. S. V. P. Varupala**, K. Sankaranarayanasamy. A heuristic method with a novel stability concept to perform parallel assembly sequence planning by subassembly detection. *Assembly Automation*, 2020.
- M. V. A. R. Bahubalendruni, A. K. Gulivindala, **S. S. V. P. Varupala**, D. K. Palavalasa. Optimal assembly sequence generation through computational approach. *Sādhanā*, 44(8), 2019.

Patent

- Smart system and method for automatic exploded view generation and assembly sequence plan optimization. Patent application filed, India.

Teaching

Graduate Teaching Assistant, Industrial and Systems Engineering 2024 – present
Wayne State University Detroit, MI
Industrial Engineering and Systems Thinking (IE 1000; designed a bike-fleet simulation game); Integrated Product Development (IE 6405); Production and Service Systems (IE 6315); Systems Engineering Capstone (SYE 7495); Lean Six Sigma Capstone; Engineering Risk and Decision Analysis; Project Management.

Lab Assistant, Hydraulics and Pneumatics, National Institute of Technology Andhra Pradesh 2021

Service and Presentations

- Presenter, ASME IDETC/CIE 2025, Anaheim, CA. Presenter, ICRAME 2020. Attendee, IWAR 2024.
- Robotics demonstrations for middle- and high-school students, Wayne State University.

Skills

Programming	Python, C++, MATLAB, C; PyTorch, PyTorch Geometric, CUDA, pybind11
Geometry and CAD	Open CASCADE (pythonocc), trimesh and Embree ray tracing, signed distance fields, CadQuery, FreeCAD API, CATIA, Fusion 360, ANSYS
Robotics	NVIDIA Isaac Sim with Newton, ROS 2, Gazebo, Intrinsic Flowstate, UR5e
Other	LaTeX, linear programming, stochastic processes, discrete-event simulation